Venkata Lakshmi Parimala Pasupuleti

Senior Backend Engineer · PropTech · AI-Powered Systems · Observability · AWS · Python · TypeScript · REST APIs · 9yr Production

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PROFESSIONAL SUMMARY

Senior backend engineer with 9+ years of production experience in PropTech and a specialization in AI-powered systems, observability, and backend ownership - the profile that maps directly to Realtor.com's News & Insights team. I have spent 8 years building and operating production backend services in the real estate and property management ecosystem (Entrata: 100K+ residential units, 1M+ property records, 99.5% uptime), which means the residential real estate domain Realtor.com serves is not new territory - it is where I have spent my career. AgentProbe is the AI proof point: a production AI evaluation and observability platform built end-to-end - structured tracing across all agent steps, 5-metric content quality engine (correctness, groundedness, accuracy, latency, token efficiency), CI-integrated regression gate blocking quality drift before deployment, and multi-provider LLM routing. This is exactly the AI-powered content quality infrastructure behind a hyperlocal content platform at scale. At Entrata I owned mission-critical Python backend systems for 7 years at 99.5% uptime - RabbitMQ async event-driven processing, 200+ PostgreSQL query optimizations, Docker microservices, GitHub Actions CI/CD, and AWS S3/EC2/RDS infrastructure. At Magizhs I was the sole technical owner: primary backend engineer on a small team (backend + frontend), delivering production to a customer in weeks. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Backend & APIs: Python (9yr), TypeScript, FastAPI, Flask, REST APIs, API Design, GraphQL (familiarity), Async Processing, RabbitMQ, Microservices, PostgreSQL, MySQL, Oracle, Redis (object caching), High-Throughput Systems, Performance Optimization, Query Optimization, Indexing, Partitioning, Stored Procedures, Distributed Systems, Reliability Engineering, Idempotency, Retry Logic, Dead-Letter Handling

AI-Powered Systems & Observability: LangChain, LangGraph, AI-Powered Content Systems, LLM Integration, AI Evaluation, Structured Logging, Distributed Tracing, Monitoring, Alerting, Observability Frameworks, CI/CD (GitHub Actions / CircleCI-compatible), Regression Testing, Content Quality Metrics, Accuracy, Groundedness, Latency, Anthropic API, OpenAI API, ChromaDB, RAG Pipelines, Agent Orchestration

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue, Lambda), Docker, Microservices, CDN concepts, Infrastructure Management, React.js, TypeScript, Linux, Bash, Git, Agile/Scrum, Technical Mentoring, Design Docs, Code Reviews, PropTech Domain (8yr residential real estate ecosystem)

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Primary Backend Owner, Small-Team Delivery

- Sole technical owner and primary backend engineer on a small cross-functional team (backend + frontend + stakeholders) - designed API architecture, built async processing layer, implemented observability and CI validation infrastructure, and delivered production deployment in weeks. Exactly the primary backend SME role Realtor.com's News & Insights team structure requires.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977) on AWS-compatible cloud infrastructure; designed observability dashboards improving research comprehension 40%; cross-functional technical delivery across 3 research pipelines; mentored 600+ students in Python, SQL, and distributed systems.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Production backend systems & PropTech domain** - owned mission-critical Python backend systems for 7 years at 99.5% uptime serving 100K+ residential units and 1M+ property records in the real estate ecosystem; RabbitMQ event-driven async processing with retry logic, idempotency guarantees, and dead-letter handling; 200+ PostgreSQL/MySQL/Oracle query optimizations through execution plan analysis, indexing, and partitioning (30% performance gain); Redis caching; AWS S3/EC2/RDS/Glue infrastructure; Docker microservices redesign (50% scalability improvement, 30% deployment time reduction); GitHub Actions CI/CD with automated regression testing.
- **Observability, reliability & operational excellence** - built structured logging, monitoring, and alerting infrastructure; authored runbooks; led incident response with RCA; built internal debugging tools reducing triage time 40%; kept bug count below 10 across 20+ production modules for 6+ consecutive months over 120+ Agile sprints - zero production incidents on owned systems. Fast feedback loops between observability signals and code improvements.
- **Technical leadership & cross-functional delivery** - wrote design documents and architecture decisions for 20+ production modules; led projects spanning multiple engineers; mentored 5+ engineers; 300+ peer code reviews; Python/Bash automation eliminating 80% of manual workload; partnered with product, compliance, and operations stakeholders across 120+ Agile sprints.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: AI-Powered Content Quality Evaluation Platform

Observability · Content Quality · CI Gate · AI Evaluation · Open Source

Python · FastAPI · LangGraph · ChromaDB · Redis · GitHub Actions · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production AI evaluation and observability platform - the infrastructure behind AI-powered content quality at scale. **Content quality engine:** 5-metric evaluation (correctness, groundedness, accuracy, latency, token efficiency) measuring AI-generated content quality before and after deployment - exactly the validation layer a hyperlocal AI content platform requires. **Observability:** structured tracing across all AI pipeline steps; monitoring and alerting on content quality drift; feedback loops from production back into evaluation suites. **CI regression gate:** GitHub Actions integration blocks quality degradation before deployment - zero regressions across 1,240+ test cases. **Multi-provider LLM routing:** Anthropic, OpenAI, Ollama with cost and latency tradeoffs. Open-source, multi-team deployable.

LeaseI: AI-Powered Document Processing Pipeline - PropTech Production System

PropTech · AI Pipeline · Multi-Agent · RAG · Production

Python · LangGraph (8-agent) · FastAPI · RAG · SQLite · MCP · Anthropic API | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production AI-powered content processing system in the real estate domain - demonstrates the agentic pipeline architecture behind AI-generated content at scale. **AI pipeline architecture:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Escalation) processes and classifies property documents with conditional routing and dynamic task delegation. **Content quality:** confidence-based routing, structured output validation at every agent boundary, citation-backed outputs with groundedness scoring. **Measured outcomes:** 93.6% precision, 87.9% F1 across 1,240 test cases - the measurement-driven approach for production AI content pipelines. Deterministic guardrails for regulated decision paths.

AI on FHIR: Full-Stack Backend - AWS Infrastructure & REST APIs

AWS · REST APIs · PostgreSQL · Observability · Production

Python · Flask · React.js · TypeScript · Docker · AWS Lambda · PostgreSQL · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack production backend on AWS - REST API backend serving LLM-powered natural-language queries as structured API calls; AWS Lambda serverless with API gateway; PostgreSQL data services; HIPAA-compliant RBAC with full audit trail and monitoring; complete observability stack. TypeScript frontend (React.js) with Docker containerization. Demonstrates backend reliability engineering, AWS infrastructure management, and production observability in a regulated environment - ~60% reduction in manual retrieval time.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017